

Are You Anonymous? Inferring Personal Information from Nonverbal Behavior Data Tracked in Immersive Virtual Reality

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ABSTRACT

The growing adoption of virtual reality (VR) technology has raised concerns regarding individuals' privacy. We explored people's ability to infer personal information about others from representations of nonverbal behavior data tracked in VR. We conducted a within-subject experiment with 53 participants to explore if they could guess personal information from gestures, spatial behavior, and a combination of both represented through neutral avatars in VR. We found that participants could not guess better than random for most features, except for age and native English speaker status. Furthermore, there were almost no differences between the various types of behavior, except for native English speaker status. We discuss how our results contradict previous research on people's identification from nonverbal behavior. Our study is the first to address people's ability to infer other people's information from nonverbal behavior in VR.

CCS CONCEPTS

• **Security and privacy** → *Human and societal aspects of security and privacy*; • **Human-centered computing** → *Empirical studies in HCI*.

KEYWORDS

virtual reality, perceived identification, nonverbal behavior, privacy

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1 INTRODUCTION

Virtual reality (VR) enables users to interact with each other through avatars responsive to their movements but not necessarily true to their appearance. This could make people assume they are anonymous enough to be safe [11]. However, research has found that VR users can be identified using machine learning (ML) algorithms with data collected from VR hardware [3, 13, 17]. Additionally, there is a long stream of research on how people can identify others or guess personal information from nonverbal behavior [5, 28]. Little research connects these two areas to explore identification in VR from nonverbal behavior by people, not algorithms.

Studying how people can guess personal features from others in VR is relevant to develop a safe virtual environment. On the one hand, if people can guess features about others by interacting in VR or watching streaming videos, VR would provide a fertile ground for information-hungry actors, even if companies secure the data collected from the hardware. On the other hand, if people cannot guess personal information from the data as algorithms easily and accurately do, there would be a disconnection between how people perceive their privacy and assess threats and risks they are exposed to, generating an illusion of anonymity in which users could act differently thinking they are anonymous [23, 30].

We explored whether people can guess personal information from representations of nonverbal behavior data collected in VR. Overall, we found that people cannot guess better than random personal information from others by watching short videos reconstructed from a VR experience. However, some exceptions open new questions in the VR privacy research area. Bridging privacy research in VR and behavior research on personal information embedded in nonverbal behaviors, our study is the first to address identification issues from VR tracking data from human understanding to our knowledge.

2 IDENTIFICATION IN VR AND NONVERBAL BEHAVIOR

Immersive VR raises privacy concerns due to its tracking system, which collects users' movement data. Miller et al. [13] and Moore et al. [14] found that ML algorithms can identify users with an accuracy higher than 90% using movement data from different activities in VR. Olade et al. [17] combined movement and eye-tracking data, achieving an even higher accuracy of 98.6%. Falk

et al. [7] provide evidence that de-anonymizing someone in VR is possible just by recording and analyzing some movement patterns.

The extent of identification could also be indirectly gauged by inferring personal information. Prior research has shown that a relatively small number of personal attributes can be combined to identify an individual uniquely. Sweeney [25] found that combining three fields of information (e.g., 5-digit ZIP, gender, date of birth) can uniquely identify more than 50% of people in the US population. Rocher et al. [19] further support this claim, showing that 99.98% of Americans can be identified in any incomplete dataset with 15 demographic attributes.

This raises new privacy questions in VR about indirect identification from tracked data in VR by pointing out the potential of a few pieces of personal information to be inferred. Nair et al. [16] found that attackers with different access levels (from end-users that could only watch the VR experience to a simulation of hardware developers that could access all the raw data) could infer up to 25 personal data attributes from someone playing an escape room game in VR deliberately designed to extract information from the user, which is more than the 15 needed to identify someone in the US [19, 25].

The potential of indirect identification includes humans as inferring actors beyond ML. Perceived identification (i.e., how people guess information from the representation of the tracked data) can yield distinct implications for privacy and data tracking in VR compared to automatic identification. Whereas automatic identification entails practical privacy risks related to how companies with access to raw data in the system can use it, how people can guess personal information from data representations influences how users feel and behave in VR and how they manage their privacy. However, little research has been done to understand people's perceptions of the identifiability potential of VR tracking data.

This study aims to understand the perceived identification of VR tracking data from a human-centered perspective. Humans collectively understand movements of the head and hands in terms of nonverbal behavior, which in VR are representations of numerical data by 3D simulations. Therefore, previous research on peoples' understanding of nonverbal behaviors benefits our approach. For example, people can infer others' personalities, behavioral habits, and other psychological phenomena from nonverbal behavior [29]. Moreover, people only need partial representations to guess information. For example, point-of-light displays have been used to prove that people can guess gender [4, 8, 24, 27] and emotions [6, 18]. Based on previous research, we explore whether people can guess personal information from nonverbal behavior data tracked in VR with the following research question (RQ1): **Can people guess other people's personal information from nonverbal behavior tracked in immersive VR?**

Identifying individuals from VR tracking data representation could be enhanced through a comprehensive study of the relationship between different types of nonverbal behavior and personal information. Thus, we additionally explore whether different types of nonverbal behavior data lead to differences in people's guesses with the second research question (RQ2): **Does the type of nonverbal behavior tracked in immersive VR affect people's accuracy in inferring personal information?** We separated the movement data gathered in VR as gesture and spatial behavior. Gestures which

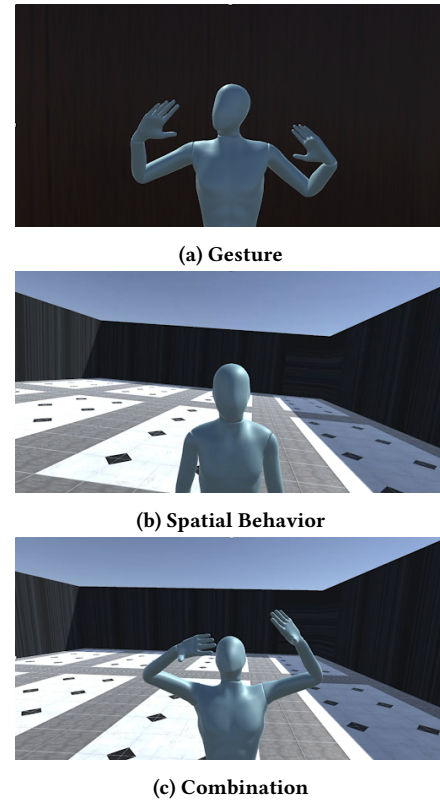


Figure 1: Example Images of the Three Conditions

refer to movements of the head and hands to convey information and facilitate communication are widely studied to enable people to guess personal information such as gender [20], ethnicity [2], and age [15]. Spatial behavior means how a person uses the space, including space occupancy, speed, and movement path, which is found to be linked to gender [21].

3 METHODS

3.1 Procedure and Participants

We conducted an IRB-approved online within-subject experiment to examine whether people can infer personal information from nonverbal behavior data tracked in VR and whether different nonverbal behavior types (gesture vs. spatial behavior vs. both together) affected their guessing. Participants were exposed to all three conditions in random order. In each condition, the participants were shown five 30-second videos representing different people's nonverbal behaviors in VR and then asked to answer short surveys guessing the personal information of the video subject. Based on the observable attributes that could be inferred from observing behavior in VR used by Nair et al. [16], we selected the following set of personal information: age, gender, ethnicity, native English speaker status, and height. We also included weight considering that body movement depends on body size [12]. Participants were asked to guess the personal information of the individual recorded

after watching a video whose order for each condition was also randomized. After each question, participants rated their confidence in their answers on a 7 Likert scale. After watching and answering the questions 15 times, demographic information was asked at the end of the survey with an open space for comments.

We recruited 53 students from Cornell University with 1 academic credit as compensation for answering an online survey. We removed from the analysis 13 participants who completed the survey in less than 15 minutes for quality control. Participants' demographics were as follows: 67.5% were women and 32.5% men; 70% were from the U.S. and 30% international; ethnicity was distributed in 50% Asian, 27.5% white, and 22.5% multi-ethnic; and 70% were native English speakers.

3.2 Stimuli

The 15 videos were reconstructions of recordings of the five researchers, which have heterogeneous demographics, performing a task in VR. The task was to describe a park in VR to another person outside VR who could not see the environment. The task of describing the virtual environment is chosen since observing and exploring a virtual environment is a common behavior of VR users when they start experiencing VR. The virtual park contained objects varying in type, size, and color, such as gazebos, fountains, bridges, and trees.

We used Oculus Quest 2, which provides 1832 x 1920 resolution per eye, supports 60, 72, and 92 Hz refresh rates, and captures six degrees of freedom (6-DOF). We decided to use hand controllers rather than hand tracking since the latter is only available when hands are visible to the headset's cameras. The environment was designed in Unity3D and recorded at 5Hz. A total of 18 variables, 6-DOF of head and two hands, were separated to reconstruct gesture and spatial behaviors. Gestures were reconstructed using all data except the x and z headset coordinates. Spatial behavior was reconstructed using the headset's x, y, and z coordinates.

The data was replayed using a default avatar to reduce pieces of human-like cues that could influence participants' guesses (Figure 1). Any aspect of additional nonverbal behaviors that can imply personal information, such as facial expressions, is not presented via the avatar. The upper body of the avatar with remapped behaviors was recorded with a screen recorder on the desktop with 1920x1080 resolution. The avatar was displayed in a default room with grids to represent spatial behavior.

To ensure consistency in the data, we cropped the videos by removing the introductory part of the video where researchers familiarized themselves with virtual movement and controllers, the parts where the researchers were listening to the questions being asked, and segments where they interacted with the virtual boundaries set up from the physical room. This process leads to a less than one-minute video duration. Finally, to ensure the same length of the video with no visual lag from the recording system, we cropped a 30-second section in the video.

4 RESULTS

To understand if the participants could infer personal information from nonverbal data (RQ1), we compared participants' guesses with random guesses as a baseline success. Defined as how well an

individual can guess personal information if they randomly guess, we generated the baseline random guess for each piece of personal information as follows: age, gender, and ethnicity were randomly selected using Cornell's distribution; native English speaker status was randomly selected based on New York distribution since it was not available for Cornell; height was randomly drawn from a normal distribution depending on gender, based on US mean and standard deviation; and weight was randomly drawn using a normal distribution based on US students' mean and standard deviation in private universities [10]. For each of the six personal features, we generated 10,000 random guesses per researcher and calculated the odds ratio (OR) within our survey population compared to our simulation. In Table 1, we reported the proportions of correct predictions for each personal information, along with the OR p -value calculated with a significance level of 0.05 [22].

We found that with gestures, participants' guesses on age (accuracy of .25) and whether a participant is a native English speaker (accuracy of .54) were both slightly better than random (accuracy of .19 for age and accuracy of .46 for native English speaker status) with p -values of .040 and .017, respectively. Participants guessed better on weight (accuracy of .14 across all 3 conditions) than random (accuracy of .09) with a p -value of .035 with combination and .040 with spatial behavior. Other than these three, participants did not guess better than chance. We did not find significant differences across types of nonverbal behavior, except gestures in guessing Native English Speaking status (p -value = .009 overall). The latter was mainly explained by its comparison to spatial behavior (p -value = .01).

5 DISCUSSION

Except for a few interesting cases, we found that people were not significantly better than chance at guessing personal information from videos reconstructed from VR data (RQ1). This finding also aligns with how they were confident about their guessings which were below 4 on average. These findings contrast with related research on identifiability from nonverbal behavior in non-VR contexts [5, 9, 27]. While we discuss the limitations of our study below, our findings point to potential new inquiries. People's perceptions about their own (in)ability to identify others from VR data representations raise questions about assessing their anonymity and privacy concerns in immersive environments.

One possible explanation for the observed discrepancy is that VR systems inadequately represent users' nonverbal behaviors. Specifically, the Oculus Quest 2 headset employed in this study tracks merely three points on users' bodies to infer the rest of the body movement. The lack of more granular data to reconstruct the avatar may have contributed to the inability of participants to guess personal information when compared to similar exercises with point-light displays, which have a higher granularity in data to represent nonverbal behavior.

Another possible explanation for these findings is that participants might not consider real-world distribution for their guessings. The assumptions about probability distributions (e.g., normal distribution for weight) and participants' way of thinking (e.g., guessing height depending on their previous guess on gender) may not be how our participants made their inference during the 30 seconds

Table 1: Comparison of the Accuracy of Participants' Guesses and Random Guesses

	Age	Gender	Ethnicity	Native English	Weight	Height
Random Guess	.19	.48	.24	.46	.09	.20
Combination	.24 (.084)	.44 (.193)	.20 (.105)	.43 (.365)	.14 (.035*)	.17 (.919)
Spatial Behavior	.20 (.764)	.48 (.729)	.20 (.154)	.40 (.086)	.14 (.040*)	.12 (.040*)
Gesture	.25 (.040*)	.48 (.837)	.18 (.034*)	.54 (.017*)	.14 (.054)	.15 (.353)

Note: We annotate the p -value in brackets and * indicates significance at the $\leq .05$ level.

video. Furthermore, there might be new underlying assumptions when participants perceive VR data influencing their guessing process, such as how much they considered the neutral avatar's characteristics to the real user. These potential explanations open a venue for new inquiries about how people interpret information from VR experiences to infer information about other people they interact with.

This finding raises a new research inquiry regarding how much nonverbal behavior data may engender significant identifiability concerns. This question is particularly pertinent to VR systems, given that the latest release by *Meta* includes eye movement and facial expression tracking. The identifiability of tracked data needs to be examined with a change of tracking system in the VR headset industry.

In addition, we need to note that although people could not infer personal information overall, we found that people could guess age and native English speaking status exclusively from gestures and weight from spatial behavior and a combination of gestures, all slightly better than random. However, the significance was barely significant, so the relevant suggestion of this finding is that future research should explore different contexts and assumptions on people's rationale for guessing.

The type of nonverbal behavior data did not generally affect participants' guessing performance (RQ 2). However, guessing native English speakers' status from gestures was better than from spatial behavior. This finding suggests that various nonverbal behaviors convey distinct personal information, underscoring the need to approach the identifiability concerns of VR tracking systems with context consideration. This aligns with the exceptional findings from RQ1 that suggest that people can guess different pieces of personal information with different nonverbal behaviors. For example, in VR tasks that entail more physical movement instead of relying heavily on gestural cues, the identifiability concerns associated with distinguishing whether a user is a native English speaker may be less pronounced.

6 IMPLICATIONS

Immersive VR raises unique identification and privacy concerns, specifically about tracked data. Our approach proposes a user-centered lens to address identification issues in VR where research on nonverbal behaviors can engage with new VR contexts. Literature has focused on how ML algorithms can identify people [3, 13, 14, 17], and our research takes an absent approach, which is how people themselves can analyze perceived identification, given that privacy is central to how users manage the boundaries of what information about themselves flows into others. Understanding how

people can infer information in different VR interactions is needed to inform developers and users about potential risks when sharing experiences and interacting with others in these new environments.

Our results about people's inability to identify users from the representation of VR-tracked data, as algorithms do, suggest that practitioners could develop mechanisms to preserve users' privacy in building the representation of the person and the environment in VR. More research is needed to identify specific ways to obfuscate identity and personal information, but our findings show that, at a general level, identification might not always be possible.

Moreover, the issue of perceived identifiability stemming from VR tracking data calls attention from the communities of human-computer interaction where a growing number of studies are leveraging nonverbal behavior data to derive human-centered design for VR [1, 26]. Researchers need to be aware of identifiability concerns associated with the data they are using to derive implications so that they can design research protocols, inform participants, and choose the open availability of datasets accordingly.

7 LIMITATIONS AND FUTURE WORK

We want to highlight a few key limitations with plans for our future studies. First, the duration of the video shown to participants was relatively short. We plan to replicate the study with different duration of videos in the future to explore the privacy risks from different extensiveness of attack. We assume that more prolonged accessibility to movement data in VR would cause greater identifiability concerns.

Moreover, more scenarios exist in which we need to explore how nonverbal behavior data in VR can leak personal information. In this study, we chose to focus on a scenario wherein individuals can view reconstructed videos of VR experiences as a starting point. We selected this scenario since two-dimensional videos of VR experiences are increasing across various video platforms in recorded or live contexts. Furthermore, reconstructing videos from default data tracked in the VR system is a relatively straightforward process. In the future, we plan to consider VR environments that provide a higher level of immersion and are associated with distinct threats and privacy concerns, whether during live or replayed interactions.

Finally, this paper represents an initial step toward comprehending the perception of anonymity through nonverbal behavioral data in VR. To advance this discussion, we will incorporate the context of VR experience and user traits. Given that specific nonverbal behaviors can facilitate the recognition of personal information by others, the specific context of VR experiences that necessitates the frequent and prominent display of such vulnerable nonverbal behaviors would enhance the identifiability of users. For example,

the VR task in Nair et al. [16] was explicitly designed to lead users to disclose information that showed different results from us unintentionally. Additionally, users' knowledge of VR will influence identifiability concerns. Users can acquire this knowledge either explicitly from the system's information design or learn implicitly through their interactions with others in VR. We intend to expand this project to encompass these two aspects and, consequently, propose design recommendations that address this issue more effectively in the future.

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